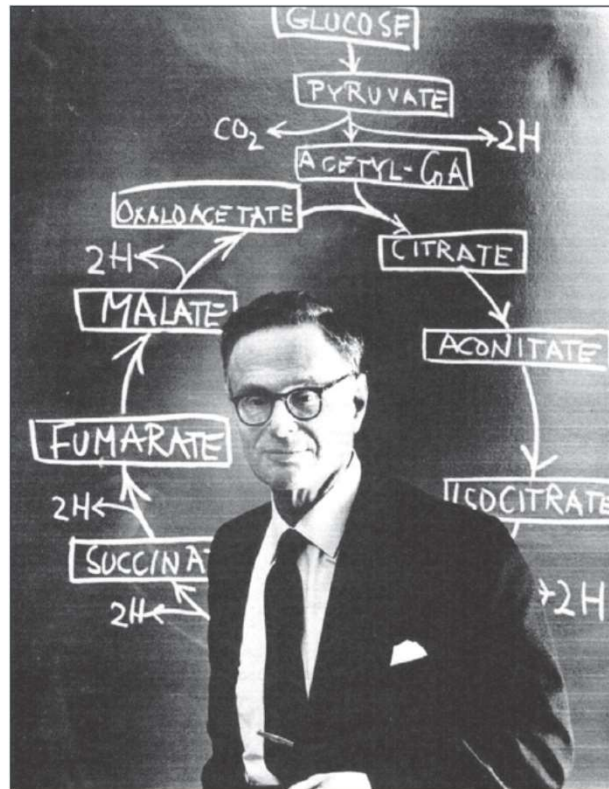


KREBS CYCLE



Hans Adolf Krebs (1900–1981)

Dr Balakrishnan. S, PhD.,
Senior Lecturer of Biochemistry

Objectives:

- At the end of this lecture, it is expected to;
 - State the Biomedical Importance of TCA cycle
 - Understand the amphibolic and anaplerotic nature of TCA cycle
 - Describe the reactions of TCA cycle in details
 - Appreciate the role of vitamins in TCA cycle
 - Understand the regulations of TCA cycle
 - Calculate the energetics of TCA cycle
 - Explain the link between TCA cycle and ETC
 - Describe the reactions of glyoxylate cycle and their importance.

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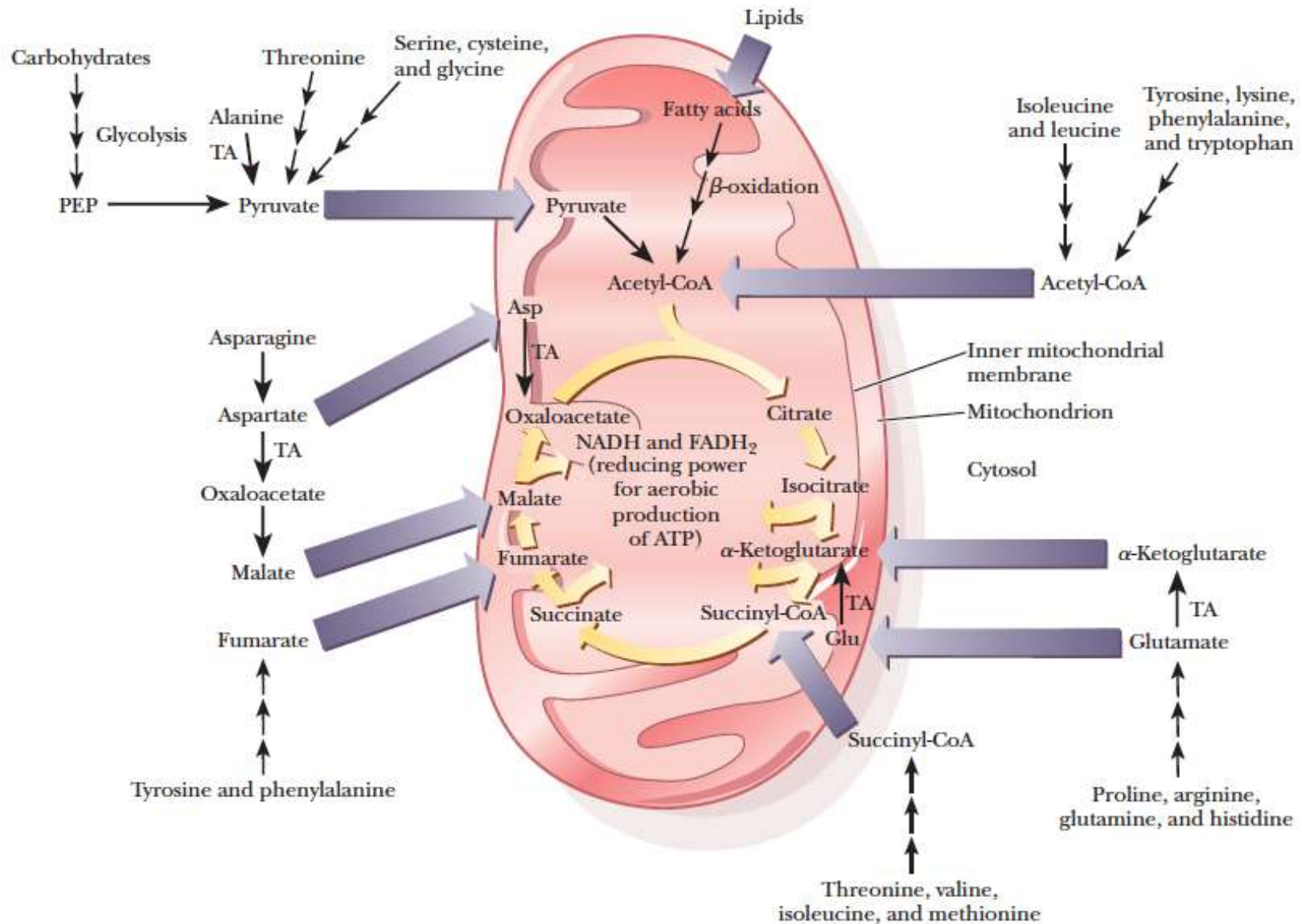
BIOMEDICAL IMPORTANCE OF CITRIC ACID CYCLE

- Final common pathway for carbohydrates, proteins and fats – through acetyl-CoA.
 - Part of **metabolic pathway** of **carbohydrates, fats and proteins** involves:
 - Convert them to **carbon dioxide** and **water**
 - Generate a form of usable energy.
- The whole process is aerobic.
 - Absence of O₂ (anoxia) or partial deficiency of O₂ (hypoxia) causes total or partial inhibition of the cycle.
- Second of three metabolic stages that are involved in **fuel molecule catabolism** and **ATP production**
 - The other two being **glycolysis** and **oxidative phosphorylation**.

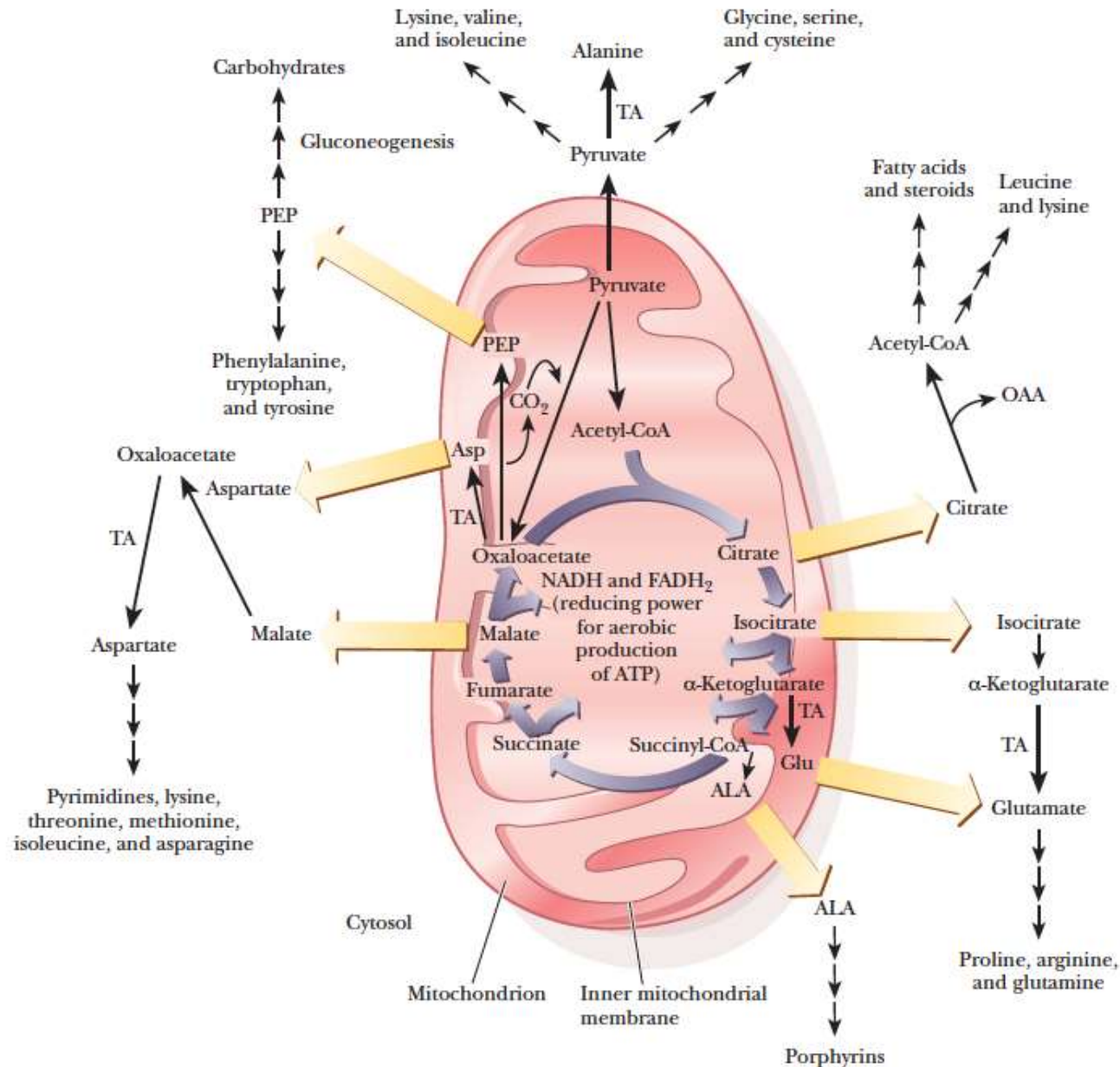
BIOMEDICAL IMPORTANCE (Cont'd)

- A series of **chemical reactions** of central importance in all living **cells**:
 - **Utilize oxygen** as part of **cellular respiration**.
- TCA cycle is amphibolic: Catabolic & Anabolic.
 - Acetyl-CoA is oxidized to CO_2 and H_2O giving out energy (catabolic role).
 - Intermediates of TCA cycle play a major role in synthesis (anabolic role) also like:
 - ✓ Heme formation,
 - ✓ Formation of non-essential amino acids,
 - ✓ FA synthesis, cholesterol and steroid synthesis.

A summary of catabolism, showing the central role of the citric acid cycle.



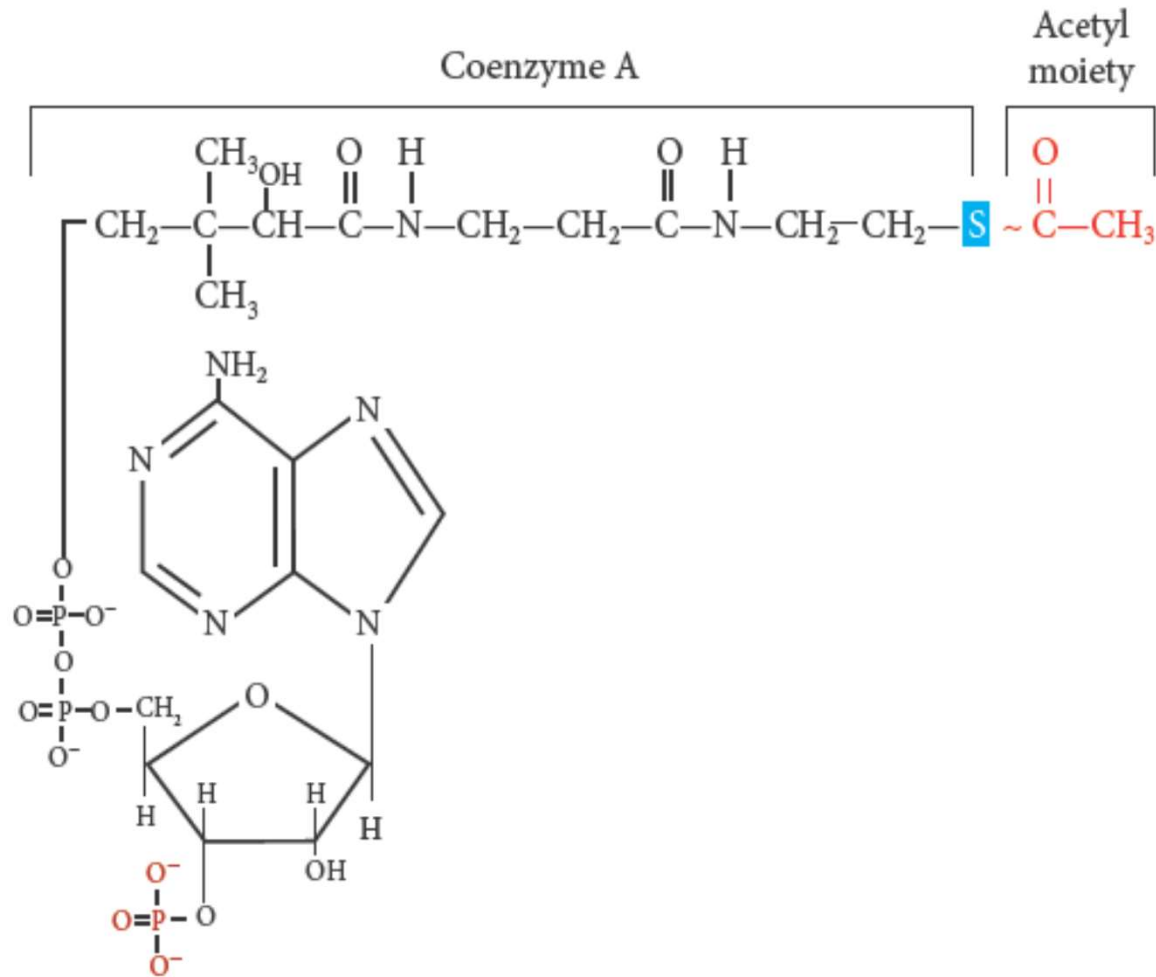
A summary of anabolism, showing the central role of the citric acid cycle.



ACETYL-CoA (Acetyl Coenzyme A, ACoA)

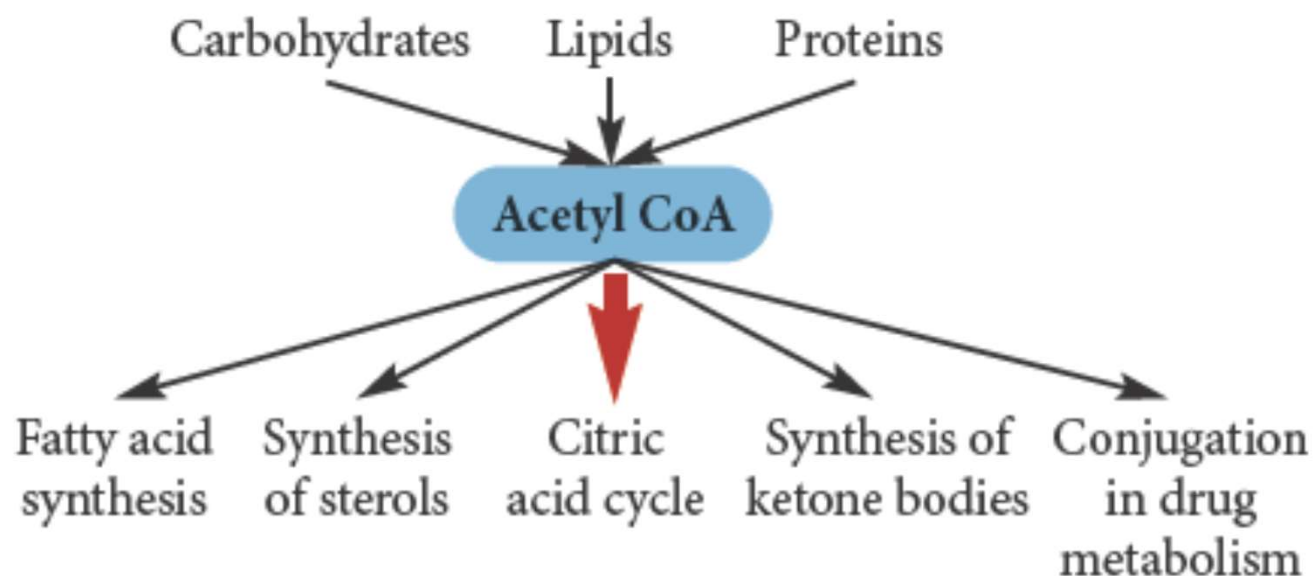
- During glycolysis, glucose → pyruvate
- Then pyruvate → ACoA by a group of enzymes known as *pyruvate dehydrogenase complex*
 - NADH is produced during those process
- In TCA cycle, two carbon atoms of the ACoA are ultimately oxidized to CO₂
- ACoA is a carrier of *acyl* groups that are derived from;
 - Pyruvate (Carbohydrates),
 - FAs (Fats) and
 - AAs (Proteins)

ACETYL COENZYME A: THE CENTRAL MOLECULE OF METABOLISM



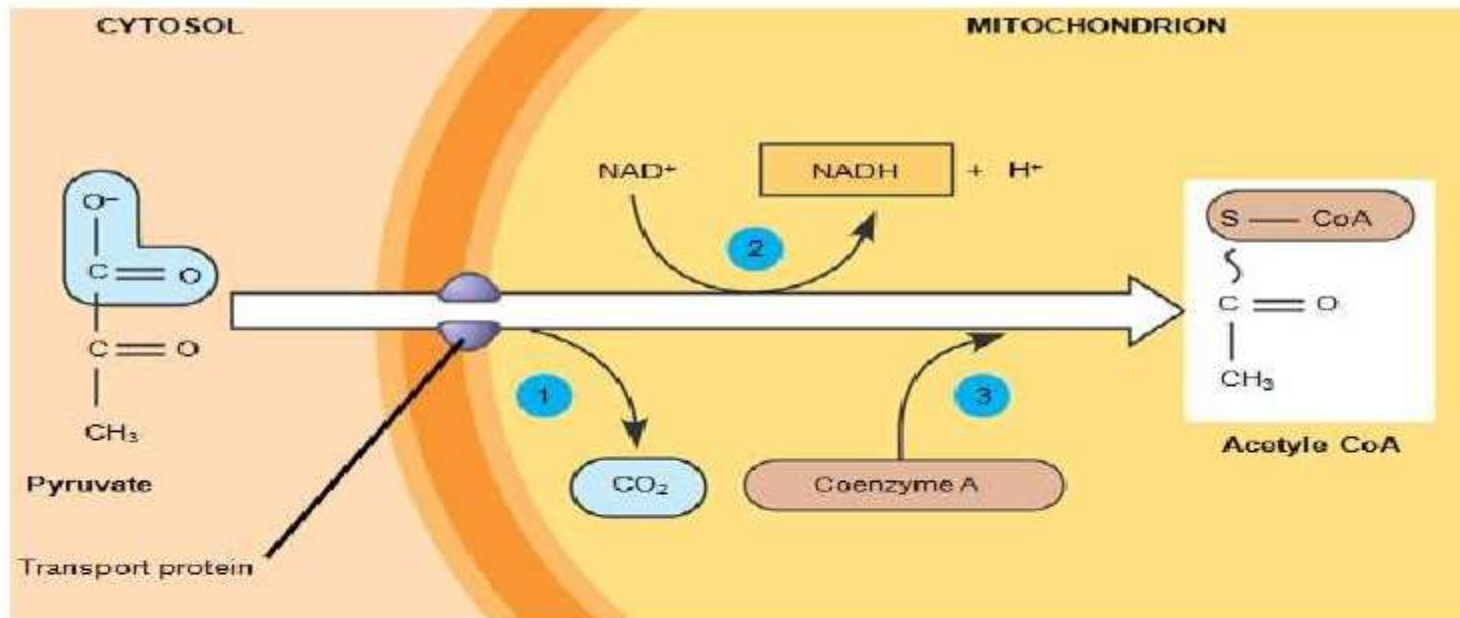
Structure of acetyl CoA

SOURCES AND FATE OF ACETYL COA



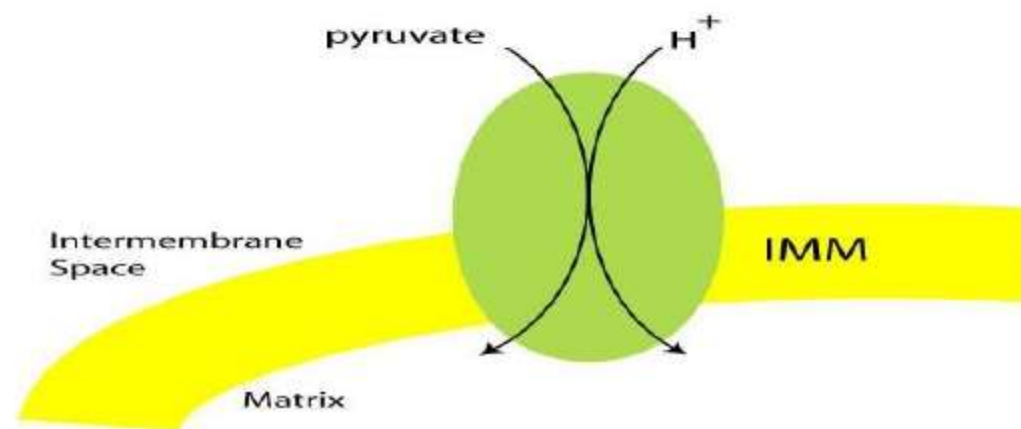
Krebs Cycle: Preparatory Step

- Pyruvic acid from glycolysis is converted to acetyl coenzyme A (A-CoA) in three main steps:
 - Decarboxylation
 - 1 carbon is removed from pyruvic acid; $3C \rightarrow 2C$ molecule
 - The lost carbon forms carbon dioxide; exhaled
 - Oxidation
 - 2 Hydrogen atoms are removed from pyruvic acid (‘oxidation’) and picked up by NAD^+
 - NAD^+ is reduced to $NADH + H^+$ (see next slide)
 - Formation of acetyl CoA – the resulting acetic acid is combined with coenzyme A, a sulfur-containing coenzyme, to form acetyl CoA (ACoA)

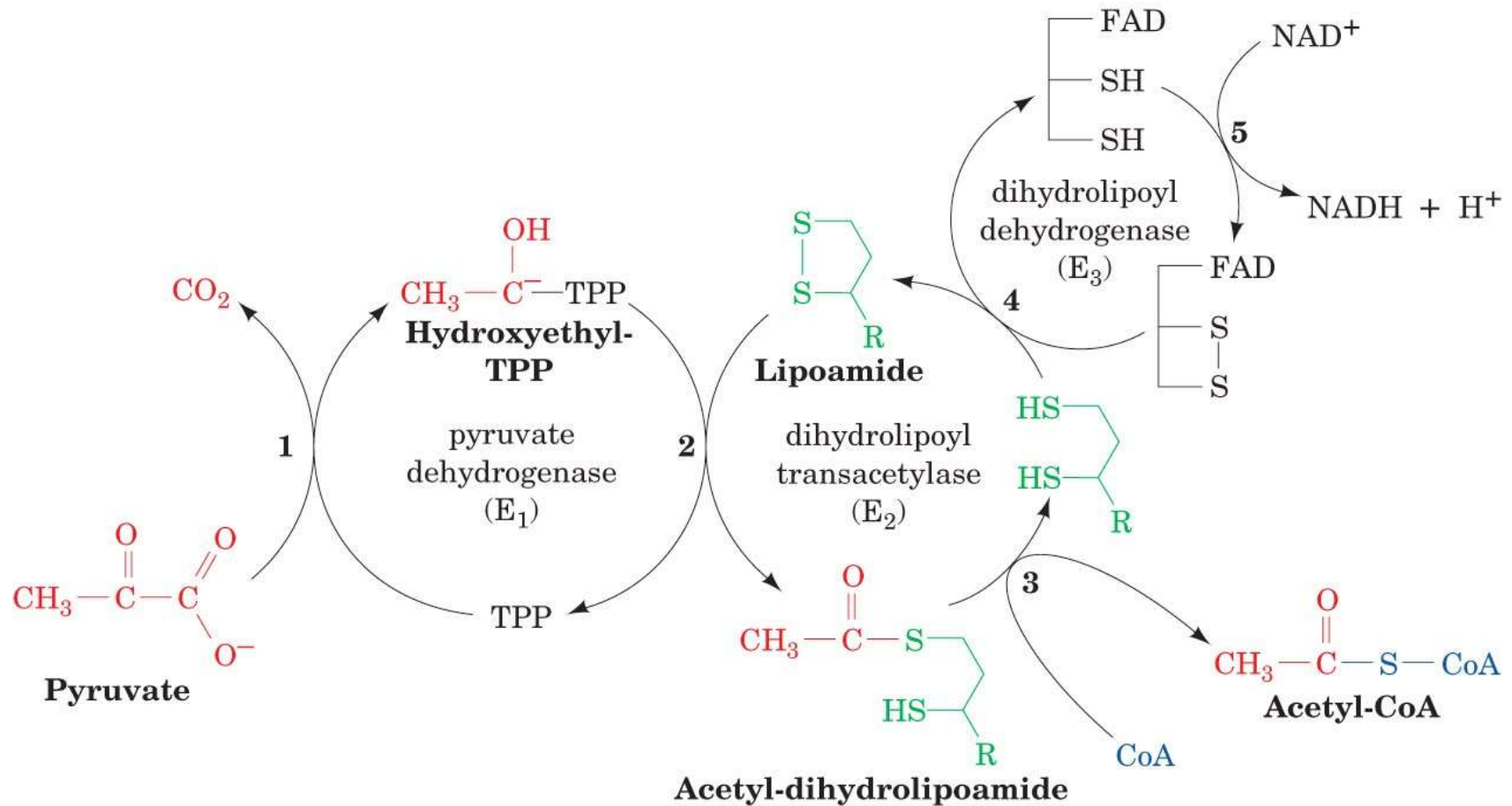


Pyruvate Symporter

Note that this symporter "costs" one proton to transport a pyruvate into the matrix of the mitochondrion

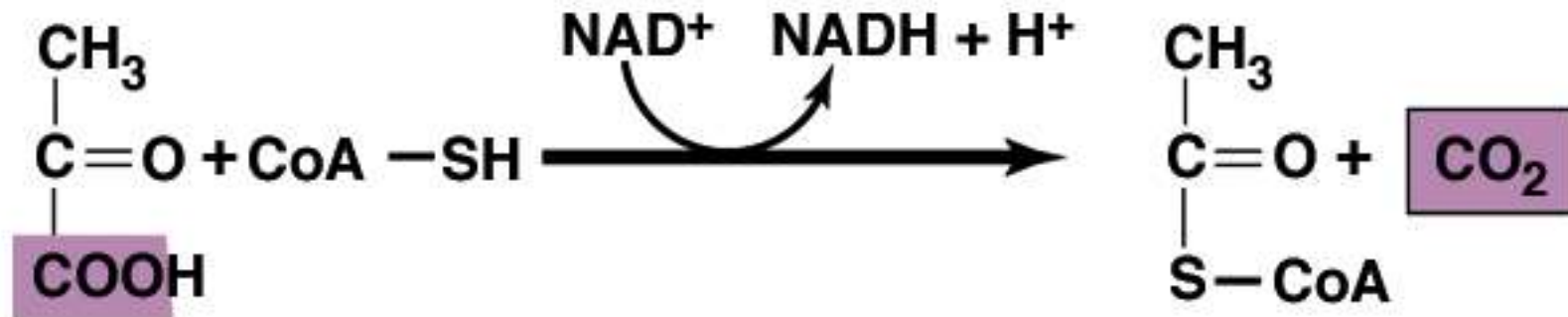


The five reactions of the PDC



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Figure 3-43



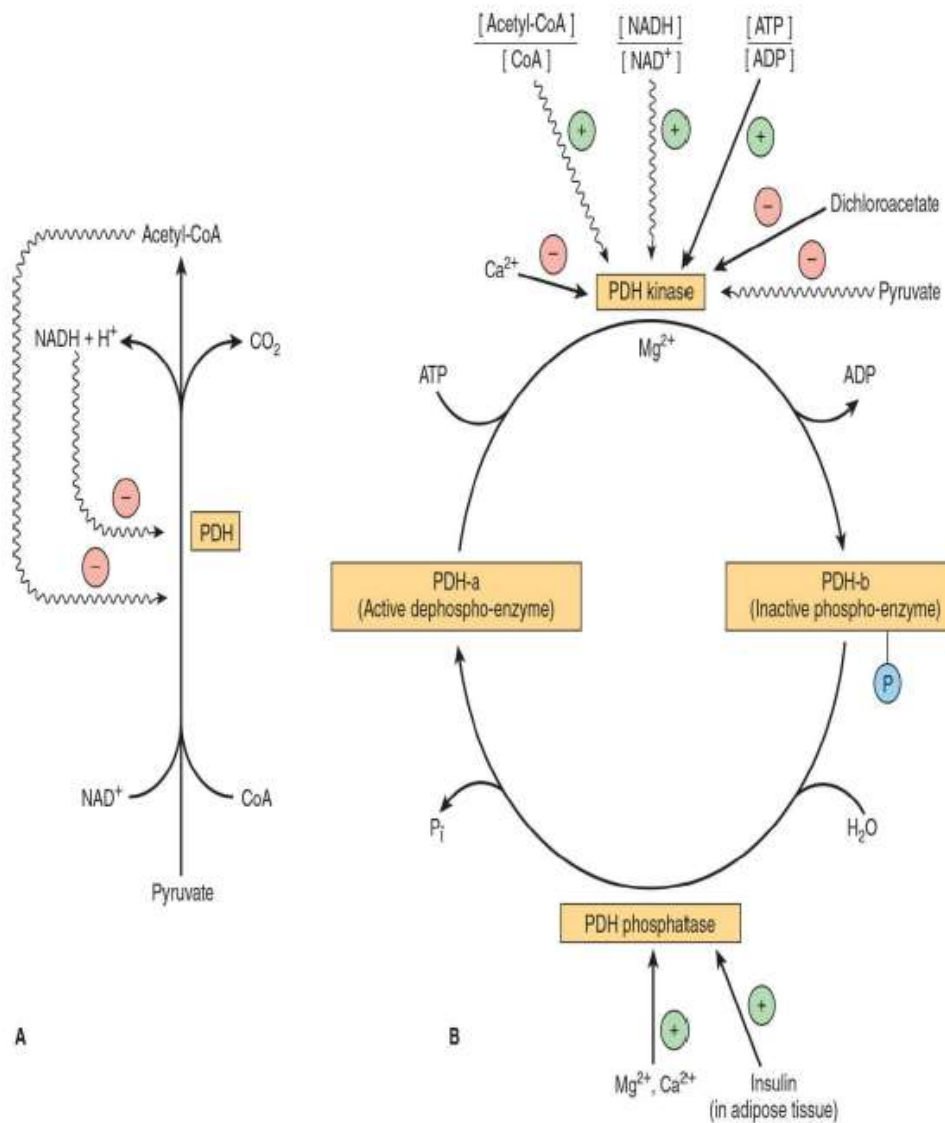
Pyruvic acid

Acetyl coenzyme A

Each transition of pyruvate to acetyl coenzyme A yields one NADH and one CO₂.

The acetyl coenzyme A then enters the Krebs cycle.

Regulation of pyruvate dehydrogenase (PDH).



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PDH Deficiency

- Caused by a deficiency of one of the components of the PDH complex
- Most common cause → mutations in the PDHA1 gene (Xp22.1; X-linked dominant trait; encodes the E1-alpha subunit).
- Clinical signs and symptoms
 - Abnormal facial shape; Feeding difficulties in infancy; Growth delay; Lethargy
 - Muscular hypotonia; Abnormality of eye movement;
 - Aplasia; Ataxia; Dysarthria; Microcephaly; Seizures; Spasticity; Tachypnea; Tremor
- Diagnosis:
 - Indicators: Early-onset neurological disease, unexplained lactic acidosis and particularly if structural cerebral abnormalities. In many cases, CSF lactate↑ disproportionately compared to blood lactate.
 - Brain MRI to check for brain damage
 - Blood test to measure levels of lactic acid or pyruvate; Blood test or urine test to analyze levels of the amino acid alanine
 - Tests to measure levels of lactic acid or pyruvate in the fluid surrounding the brain and spinal cord (cerebrospinal fluid)
- Treatment options typically include supplementing cofactors including carnitine, thiamine, and lipoic acid

A case file: Pyruvate Dehydrogenase Deficiency

- A patient with Pyruvate Dehydrogenase Deficiency presents with symptoms such as developmental delays, neurological abnormalities, muscle weakness, seizures, and lactic acidosis.
- Biochemical changes:
 - ↻ ↑lactate and pyruvate in the blood and cerebrospinal fluid.
- Diagnosis:
 - Genetic testing and
 - Biochemical analysis of lactate and pyruvate levels.
- Treatment:
 - Dietary modifications,
 - Supplements like thiamine and lipoic acid, and
 - Management of seizures and neurological symptoms.
 - Regular monitoring of lactate and pyruvate levels - is crucial for managing effectively.

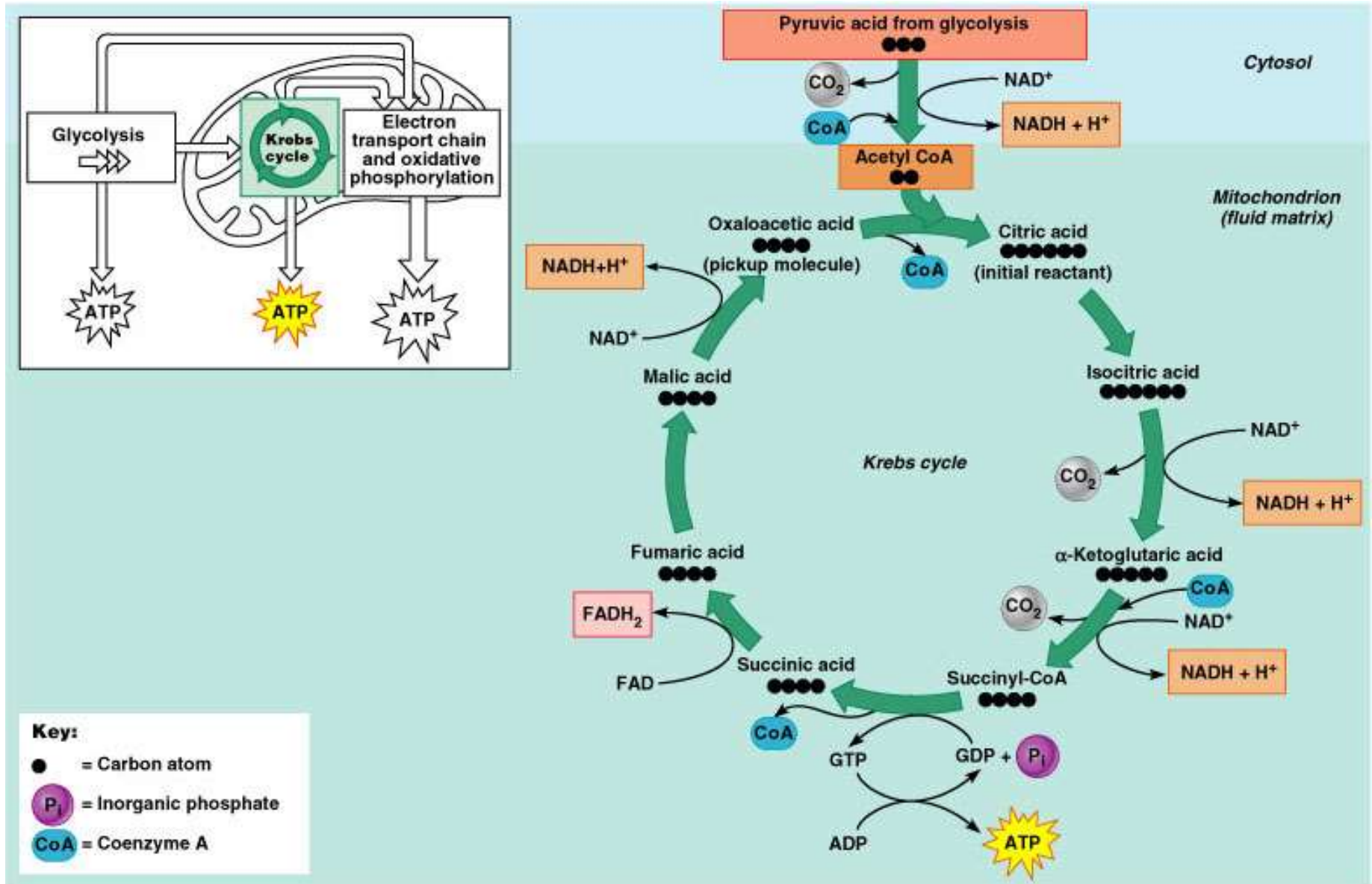
Krebs Cycle

- Discovered by **Sir Hans Krebs** in 1953 - awarded Nobel Prize in Medicine
- An eight-step cycle in which *each* acetic acid is decarboxylated and oxidized, generating:
 - Three molecules of $\text{NADH} + \text{H}^+$ (ox/red)
 - One molecule of FADH_2 (ox/red)
 - Two molecules of CO_2 (decarboxylation)
 - One molecule of ATP (substrate level phosphorylation)
- For each molecule of glucose entering glycolysis, two molecules of acetyl CoA enter the Krebs cycle
 - Remember, 1 - 6C Glucose $\rightarrow$ 2 - 2C acetyl coenzyme A (A-CoA)

TABLE 3–9 Characteristics of the Krebs Cycle

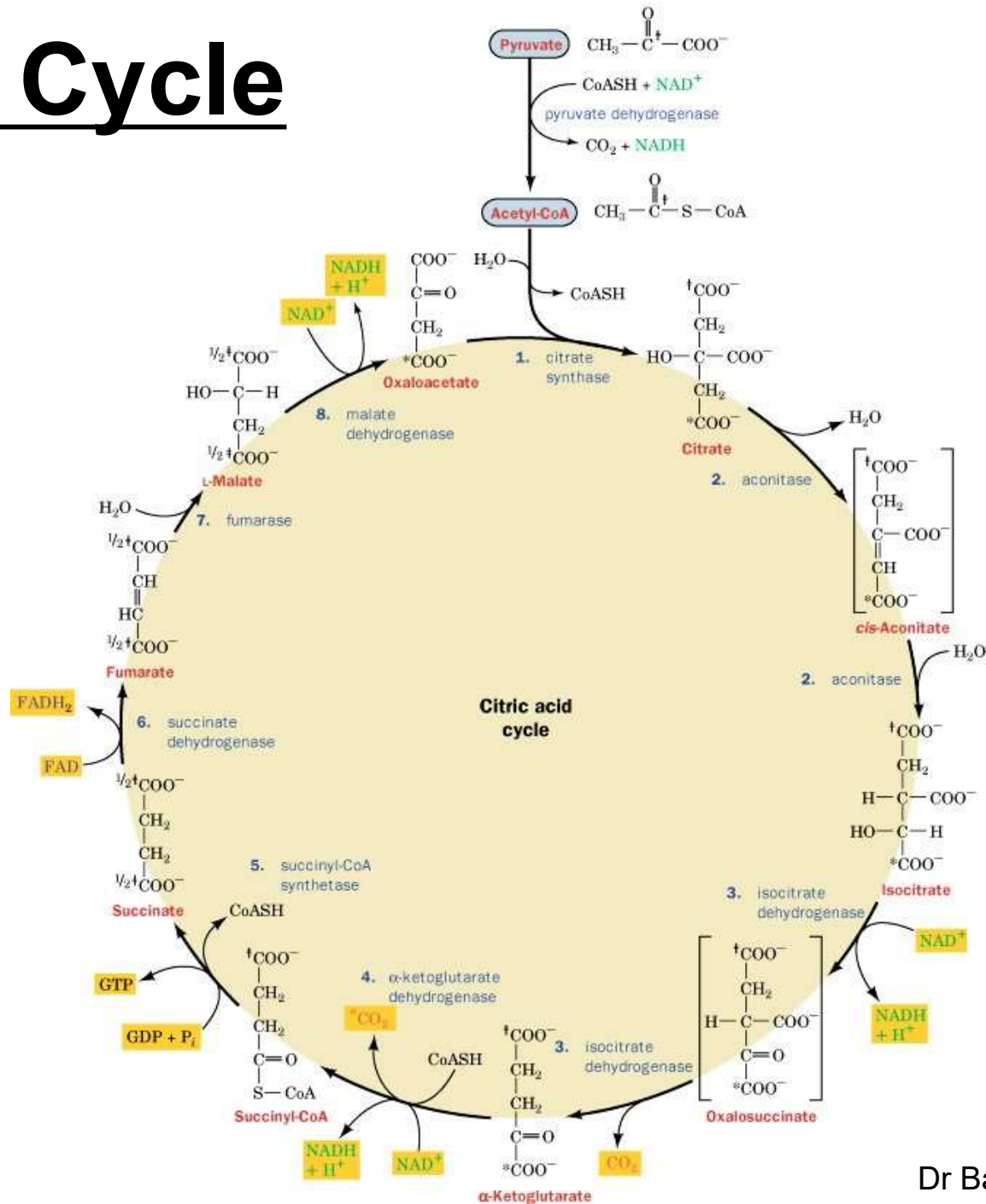
Entering substrate	Acetyl coenzyme A—acetyl groups derived from pyruvate, fatty acids, and amino acids Some intermediates derived from amino acids
Enzyme location	Inner compartment of mitochondria (the mitochondrial matrix)
ATP production	1 GTP formed directly, which can be converted into ATP Operates only under aerobic conditions even though molecular oxygen is not used directly in this pathway
Coenzyme production	3 NADH + 3 H ⁺ and 2 FADH ₂
Final products	2 CO ₂ for each molecule of acetyl coenzyme A entering pathway Some intermediates used to synthesize amino acids and other organic molecules required for special cell functions
Net reaction	$\text{Acetyl CoA} + 3 \text{ NAD}^+ + \text{FAD} + \text{GDP} + \text{P}_i + 2 \text{ H}_2\text{O} \longrightarrow$ $2 \text{ CO}_2 + \text{CoA} + 3 \text{ NADH} + 3 \text{ H}^+ + \text{FADH}_2 + \text{GTP}$

Krebs Cycle - Overview



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TCA Cycle

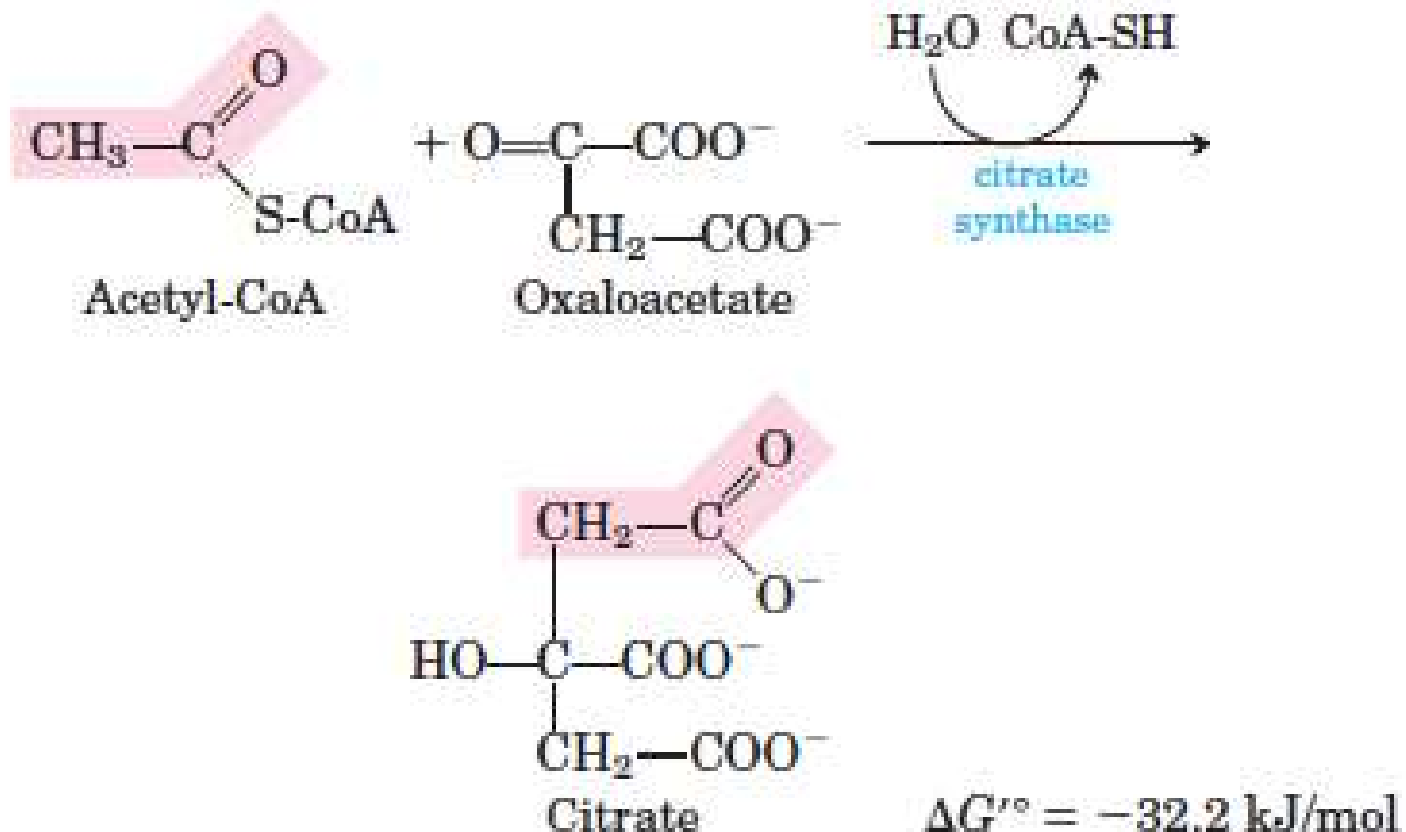


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REACTIONS OF THE CITRIC ACID CYCLE

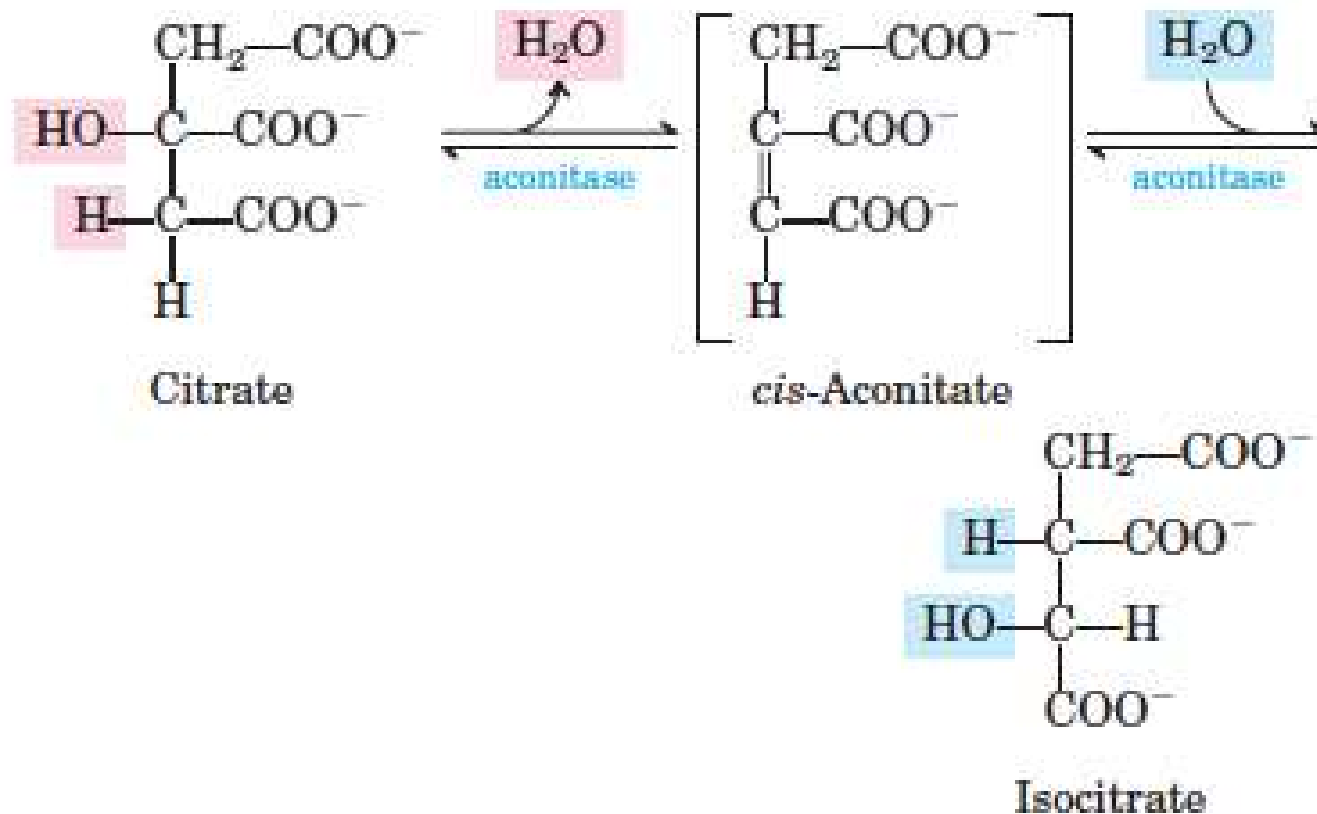
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1. Formation of Citrate



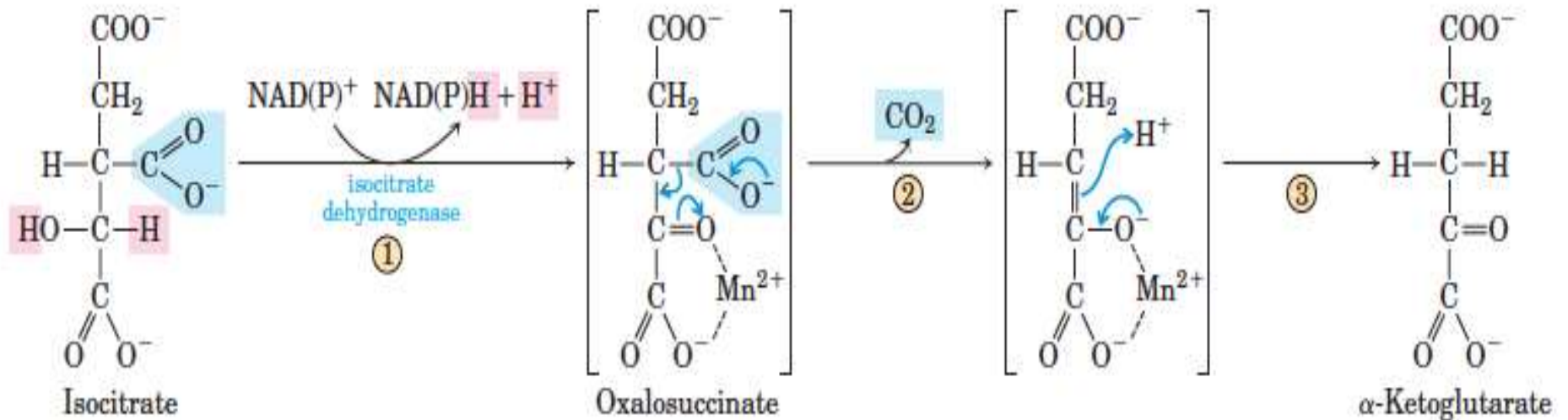
- CITRATE SYNTHASE catalyses the *condensation* of ACoA with oxaloacetate to form **citrate**. The reaction is reversible, but is a main regulatory point.
- A low NAD^+/NADH ratio and succinyl- CoA inhibits its activity

2. Formation of Isocitrate via cis-Aconitate



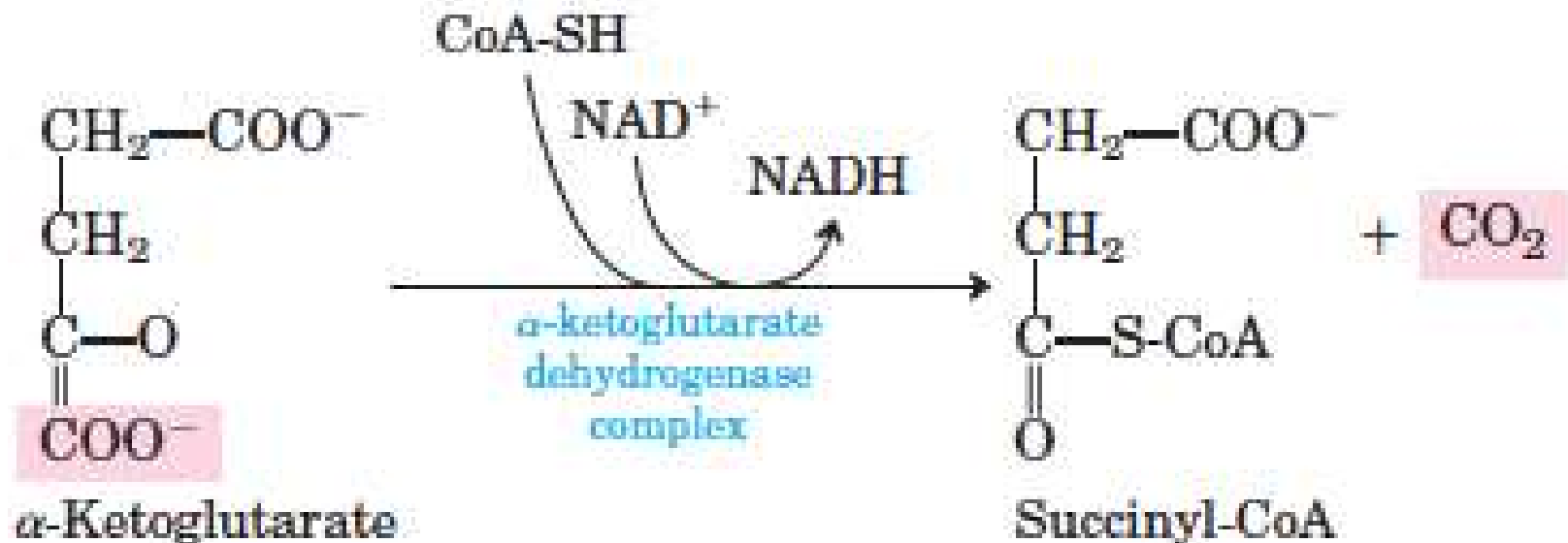
- ACONITASE reversibly catalyses the conversion of citrate to **isocitrate**

3. Oxidation of Isocitrate to α -Ketoglutarate and CO_2



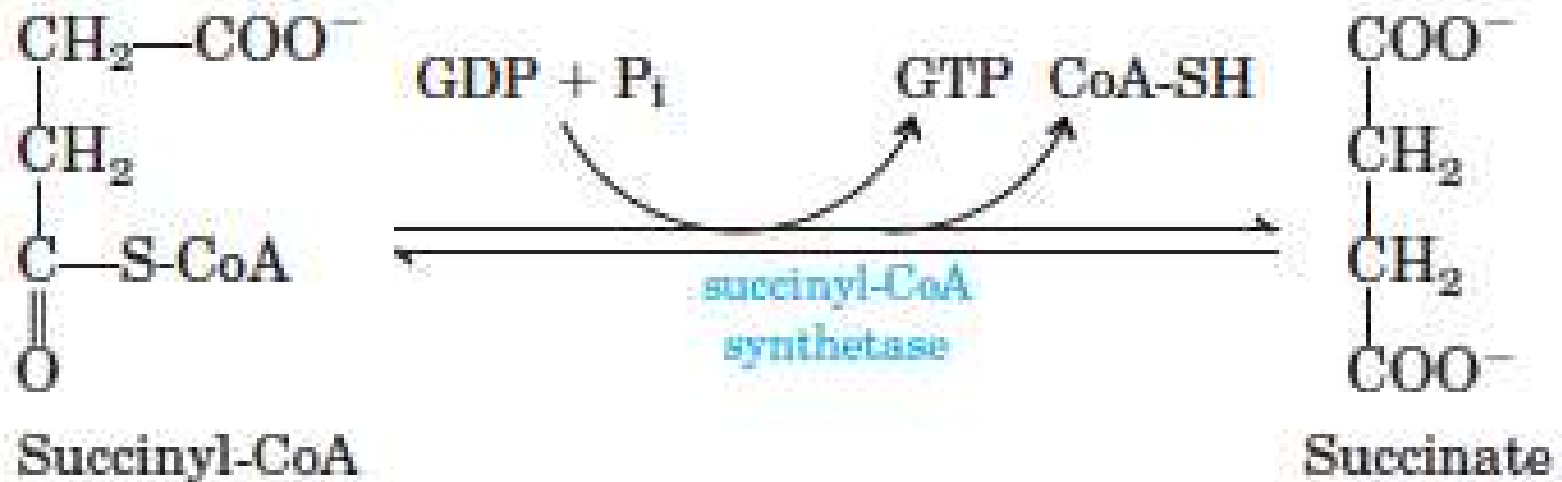
- ISOCITRATE DEHYDROGENASE (IDH_3) oxidatively decarboxylates isocitrate to **α -ketoglutarate** (2-oxyglutarate).
 - In this process NAD^+ s reduced to NADH and CO_2 is released.
 - Isocitrate dehydrogenase is inhibited by ATP and NADH and activated by ADP and NAD^+
- IDH_1 and IDH_2 found outside mitochondria use NADP^+

4. Oxidation of α -Ketoglutarate to Succinyl-CoA and CO_2



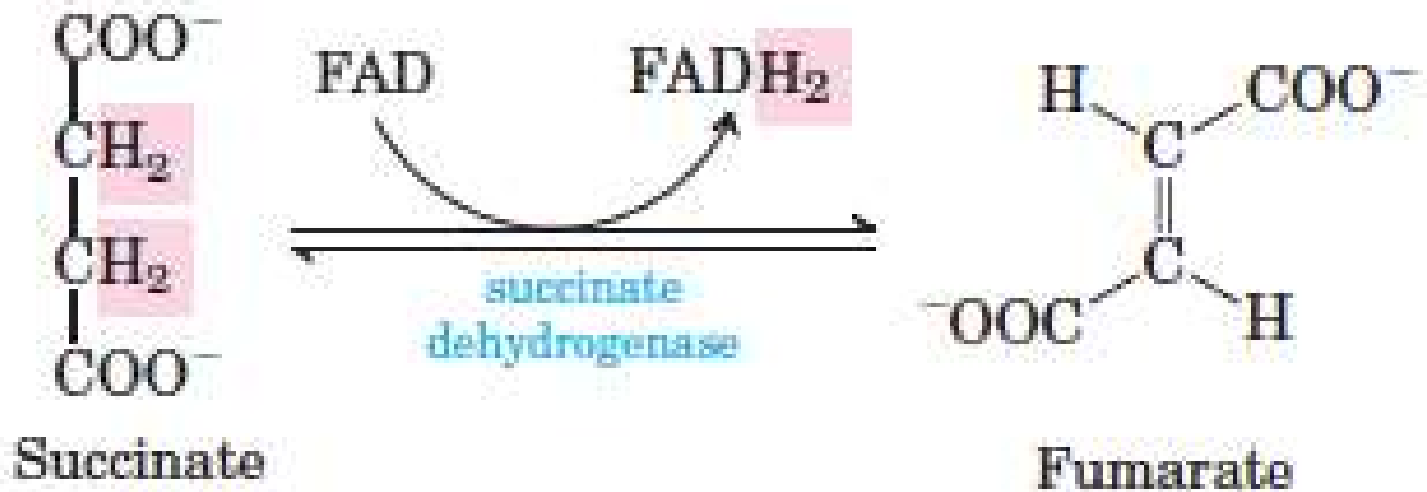
- α -KETOGLUTARATE DEHYDROGENASE produces **succinyl-CoA** from α -ketoglutarate and coenzyme A.
- Another NAD^+ is reduced to NADH and CO_2 is released.
- Both NADH and succinyl-CoA inhibit the enzyme complex, *2-oxygluturate dehydrogenase complex*.

5. Conversion of Succinyl-CoA to Succinate



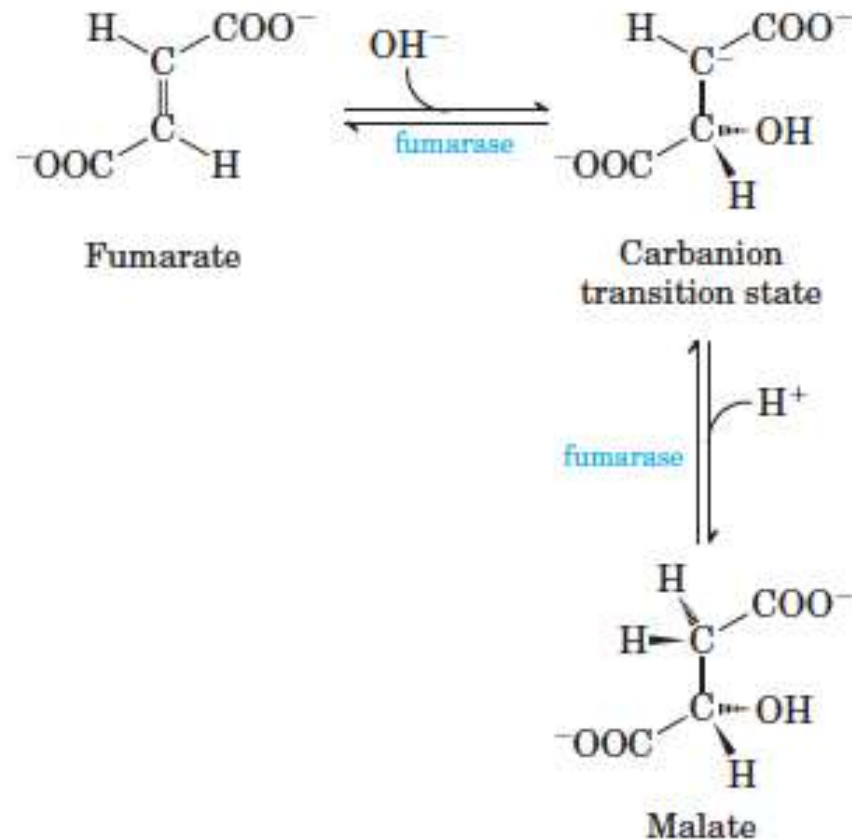
- SUCCINYL-CoA SYNTHETASE converts succinyl-CoA to **succinate**.
- GDP is converted to GTP during this step.
- This is the only step in TCA (citric acid cycle) that involves **substrate-level phosphorylation**.

6. Oxidation of Succinate to Fumarate



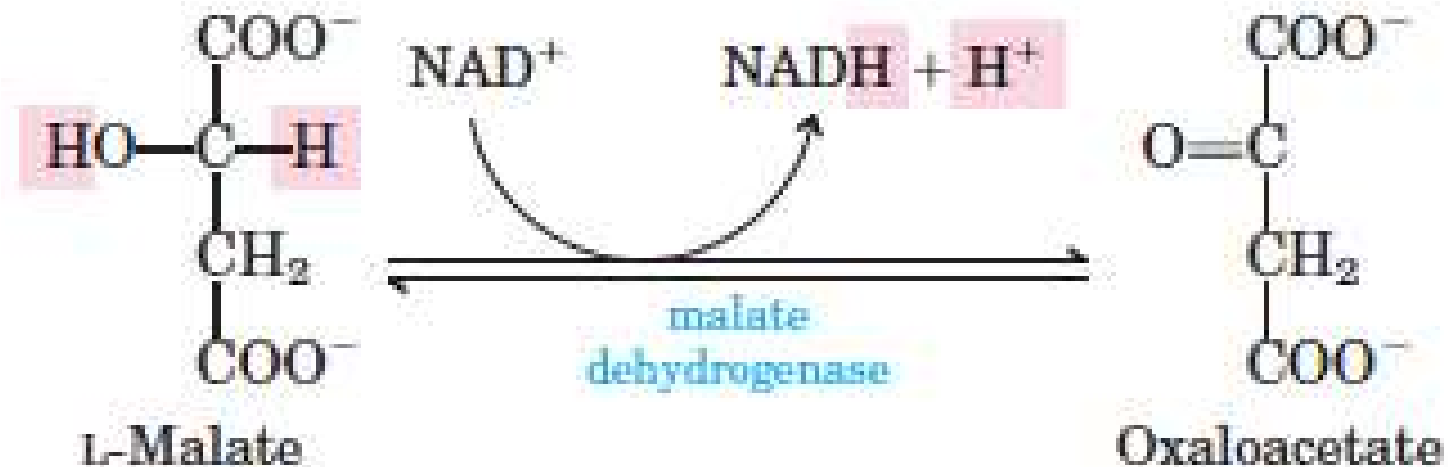
- **SUCCINATE DEHYDROGENASE SYNTHETASE** oxidizes succinate to **fumarate**.
- This enzyme is membrane bound in the mitochondria and transfers two H^+ to FAD to form FADH_2 .
- It is inhibited by oxaloacetate.

7. Hydration of Fumarate to Malate

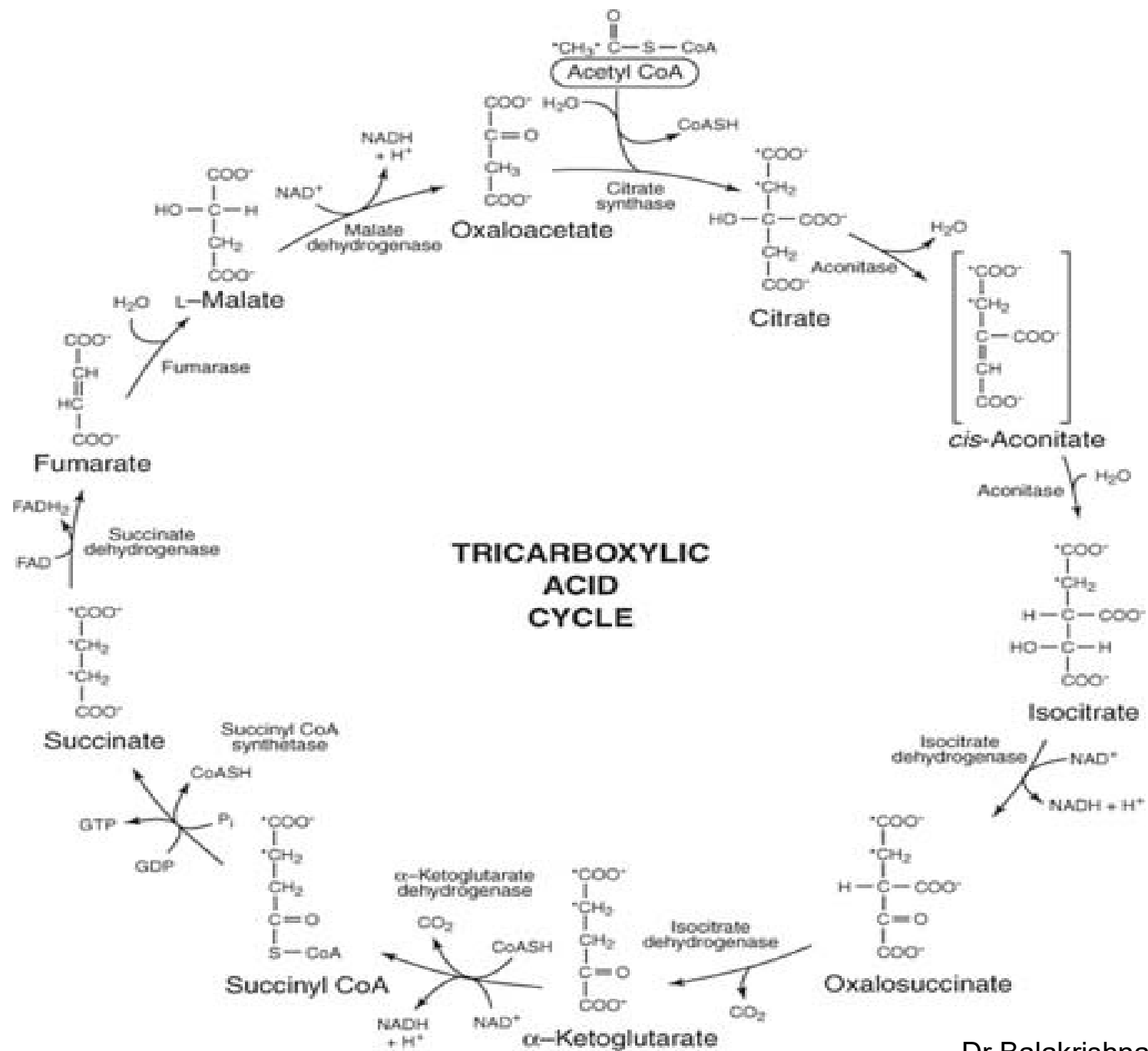


- FUMARATE HYDRATASE (fumarase) reversibly hydrates fumarate to form **malate**.

8. Oxidation of Malate to Oxaloacetate



- MALATE DEHYDRGENASE forms **oxaloacetate** and one more NADH from malate to complete the cycle



Summary of Citric acid cycle

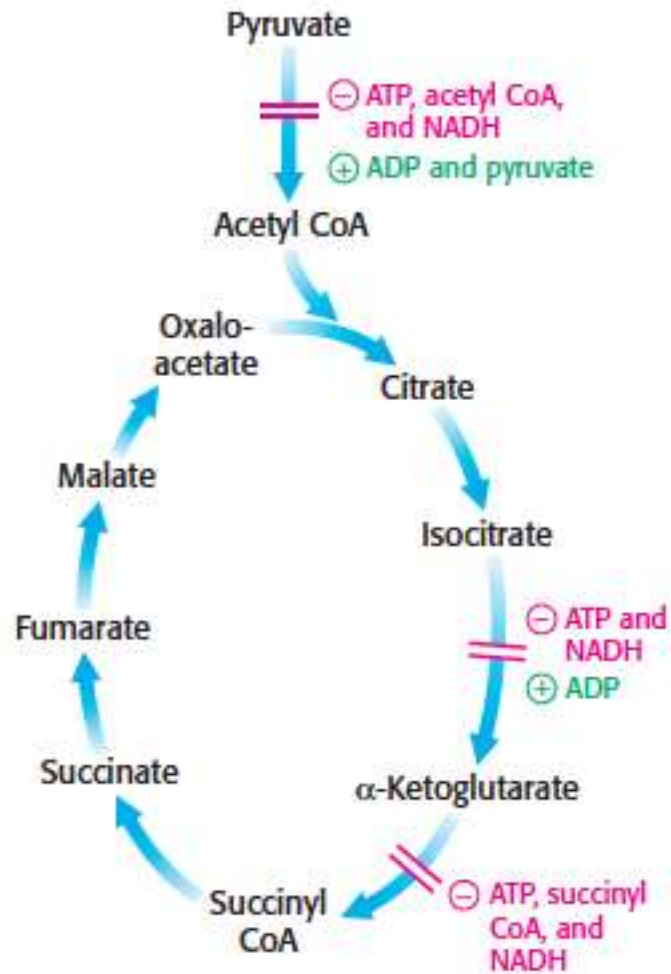
Step	Reaction	Enzyme	Prosthetic group	Type*
1	Acetyl CoA + oxaloacetate + H ₂ O → citrate + CoA + H ⁺	Citrate synthase		a
2a	Citrate ⇌ <i>cis</i> -aconitate + H ₂ O	Aconitase	Fe-S	b
2b	<i>cis</i> -Aconitate + H ₂ O ⇌ isocitrate	Aconitase	Fe-S	c
3	Isocitrate + NAD ⁺ ⇌ α-ketoglutarate + CO ₂ + NADH	Isocitrate dehydrogenase		d + e
4	α-ketoglutarate + NAD ⁺ + CoA ⇌ succinyl CoA + CO ₂ + NADH	α-ketoglutarate dehydrogenase complex	Lipoic acid, FAD, TPP	d + e
5	Succinyl CoA + P _i + ADP ⇌ succinate + ATP + CoA	Succinyl CoA synthetase		f
6	Succinate + FAD (enzyme-bound) ⇌ fumarate + FADH ₂ (enzyme-bound)	Succinate dehydrogenase	FAD, Fe-S	e
7	Fumarate + H ₂ O ⇌ L-malate	Fumarase		e
8	L-Malate + NAD ⁺ ⇌ oxaloacetate + NADH + H ⁺	Malate dehydrogenase		e

*Reaction type: a, condensation; b, dehydration; c, hydration; d, decarboxylation; e, oxidation; f, substrate-level phosphorylation.

ROLE OF VITAMINS IN TCA CYCLE

- Five B-vitamins are associated with TCA cycle essential for yielding energy.
 - **Riboflavin:** In the form of FAD - a cofactor for succinate dehydrogenase enzyme.
 - **Niacin:** In the form of NAD - the electron acceptor for isocitrate dehydrogenase, α -ketoglutarate dehydrogenase, and malate dehydrogenase.
 - **Thiamine:** As “TPP” - required as coenzyme for decarboxylation in the α -ketoglutarate dehydrogenase reaction.
 - **Lipoic acid:** As coenzyme for α -ketoglutarate dehydrogenase reaction.
 - **Pantothenic acid:** As part of coenzyme A, the cofactor attached to “active” carboxylic acid residues such as acetyl-CoA and succinyl-CoA.

Control of the citric acid cycle.



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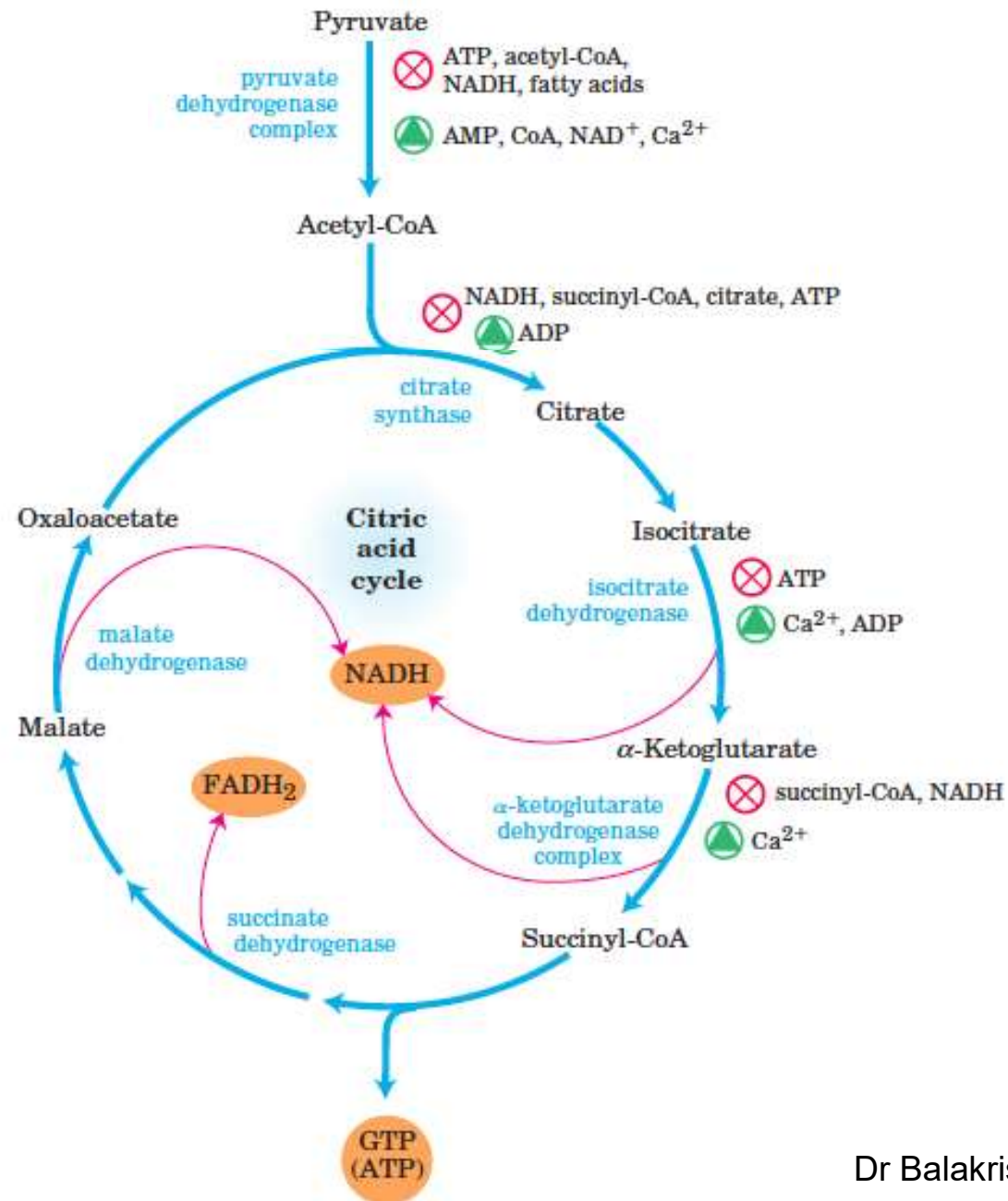
REGULATION OF TCA CYCLE

- Many of the enzymes in the TCA cycle are regulated by **negative feedback** from ATP when the **energy charge** of the cell is high.
 - Such enzymes include the **pyruvate dehydrogenase complex** that synthesises the acetyl-CoA needed for the first reaction of the TCA cycle.
 - Also the enzymes **citrate synthase, isocitrate dehydrogenase and alpha-ketoglutarate dehydrogenase**, that catalyze the first three steps of the TCA cycle, are inhibited by high concentrations of ATP.
- This regulation ensures that the TCA cycle will not oxidise excessive amounts of pyruvate and acetyl-CoA when ATP in the cell is plentiful.
 - This type of negative regulation by ATP is by an **allosteric mechanism**

REGULATION contd.

- Several enzymes are also negatively regulated when the level of reducing equivalents in a cell are high (**high ratio of NADH/NAD⁺**).
 - This mechanism for regulation is due to **substrate inhibition** by NADH of the enzymes that use NAD⁺ as a substrate.
 - This includes both the entry point enzymes **pyruvate dehydrogenase and citrate synthase**.

Regulation of the Citric Acid Cycle



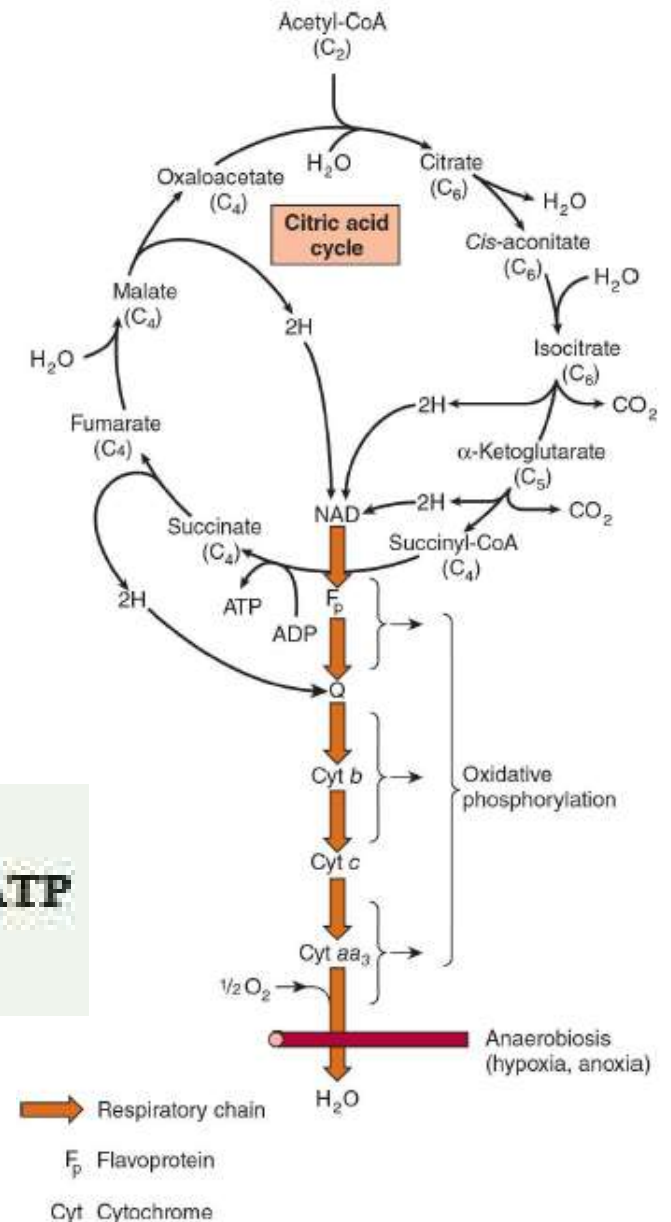
ENERGETICS OF TCA & ITS LINK TO ETC

- The rate of utilization of the ACoA in the cycle depends on the energy status within the mitochondria.
 - Under conditions of high energy, the concentrations of NADH, FADH₂ and ATP are high, and those of NAD⁺ and FAD⁺ are low → The reoxidation of NADH and FADH₂ occurs in the electron transport chain and is necessary for the cycle to continue.

Oxidative decarboxylation of PA

1. Pyruvate dehydrogenase complex
(2 NADH → 2 NAD⁺)

+ 6 ATP



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ENERGETICS OF TCA

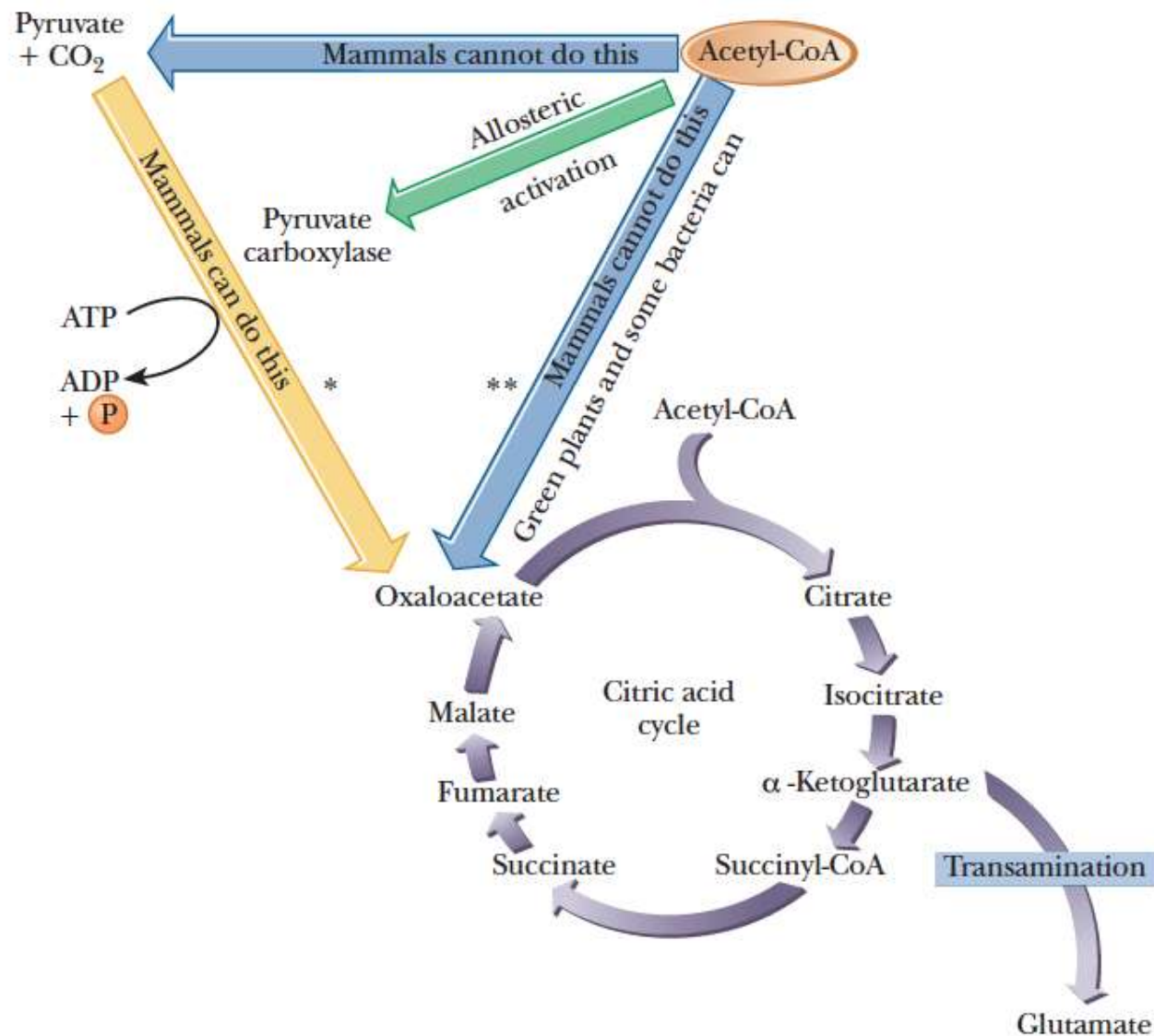
TCA Cycle

- | | |
|--|---------|
| 1. Isocitrate dehydrogenase
2 (NADH $\rightarrow$ NAD ⁺) | + 6 ATP |
| 2. α -Ketoglutarate dehydrogenase complex
2 (NADH $\rightarrow$ NAD ⁺) | + 6 ATP |
| 3. Substrate level phosphorylation
succinate thiokinase
2 GTP or 2 ITP $\rightarrow$ 2 ATP | + 2 ATP |
| 4. Succinate dehydrogenase
2(FAD.H ₂ $\rightarrow$ FAD) | + 4 ATP |
| 5. Malate dehydrogenase
2(NADH $\rightarrow$ NAD ⁺) | + 6 ATP |

A case file: Citrate Synthase Deficiency

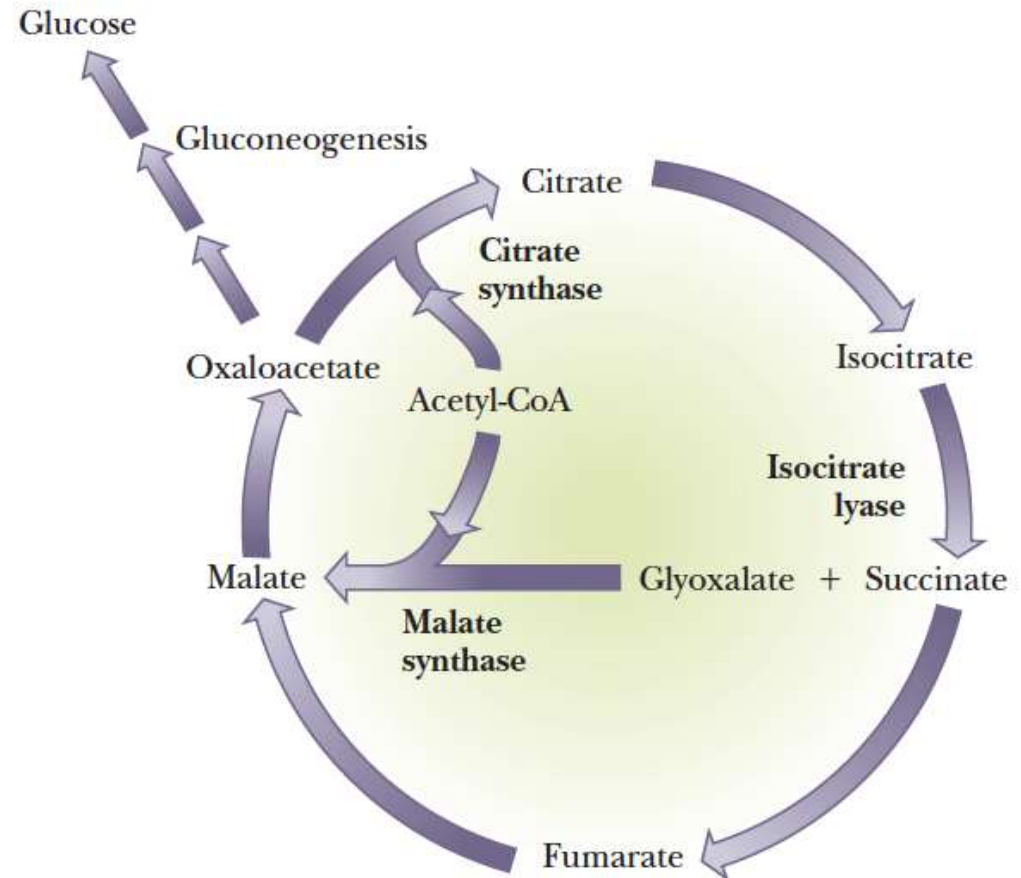
- A patient with citrate synthase deficiency presents with symptoms such as fatigue, muscle weakness, and metabolic abnormalities.
- Biochemical changes:
 - Reduced activity of citrate synthase,
 - Impaired energy production.
 - Altered lipid metabolism
 - Altered amino acids profiles
 - Impaired gluconeogenesis
- Treatment:
 - Focus on addressing metabolic adaptations and supporting mitochondrial function to optimize energy production and overall cellular metabolism.

How mammals keep an adequate supply of metabolic intermediates.

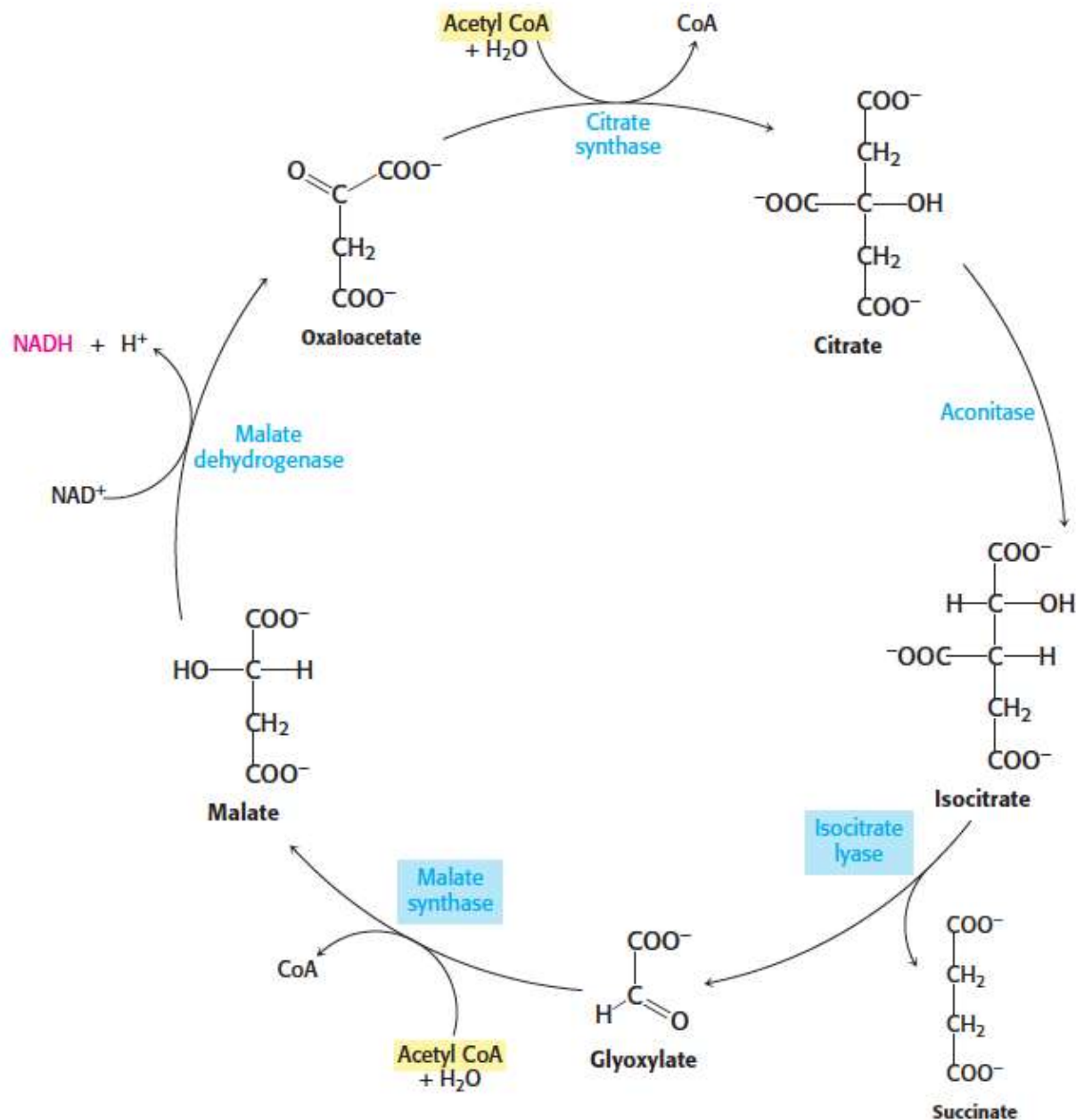


The glyoxylate cycle

- Takes place in many plants and bacteria
- In plants, it takes place in glyoxysomes
- Converts 2C acetyl units into 4C units (succinate) for energy production and biosyntheses including the synthesis of glucose
- Similar to the TCA cycle but bypasses the two decarboxylation steps of the cycle
- Two acetyl CoA's enter per turn of the glyoxylate cycle, compared with one in the citric acid cycle
- Powers seedling growth until the cell can begin photosynthesis

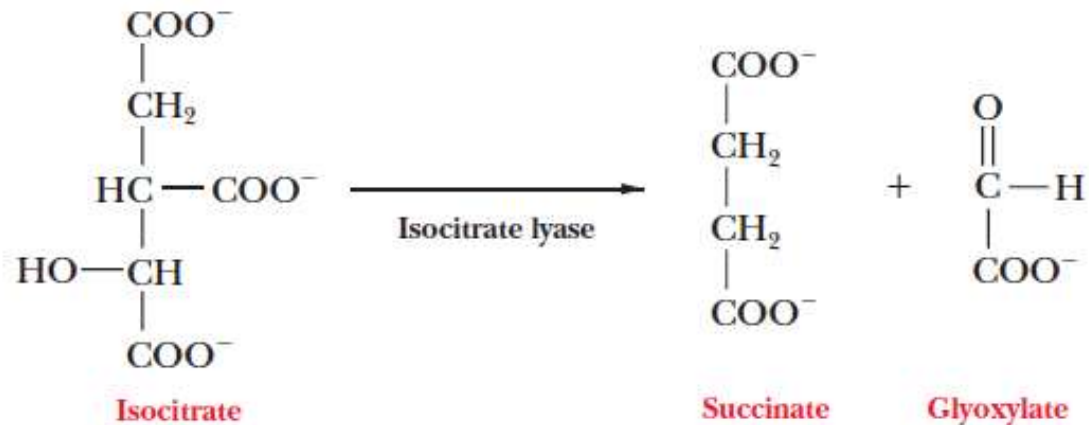


The glyoxylate pathway

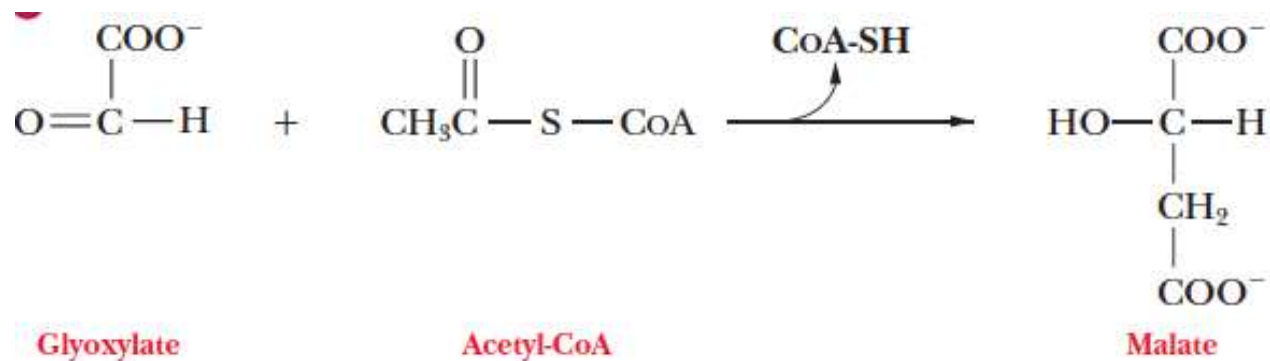


The glyoxylate cycle

The unique reactions of the glyoxylate cycle

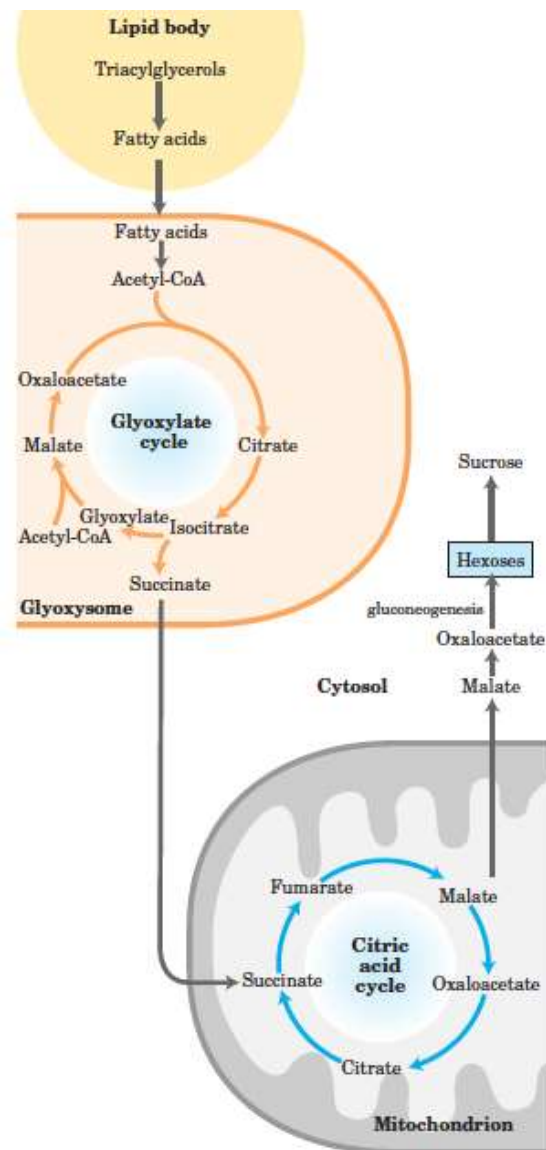


The conversion of isocitrate to glyoxylate and succinate



The reaction of glyoxylate with acetyl-CoA to produce malate

Relationship between the glyoxylate and citric acid cycles.



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Read more...



THANK YOU...

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